

- 1) TOGGLE LED WITH SOME DELAY USING PUSH PULL CONFIGURATION USING OUTPUT PIN

```
#include "stm32f407xx.h"
```

```
void delay(void){
```

```
    for(uint32_t i = 0; i < 5000000 ; i++);
```

```
}
```

```
int main(void){
```

```
    GPIO_Handle_t Gpioled;
```

```
    Gpioled.pGPIOx = GPIOD; //pd12 led4 so port d
```

```
    Gpioled.GPIO_PinConfig.GPIO_PinNumber = GPIO_PIN_NO_12;
```

```
    Gpioled.GPIO_PinConfig.GPIO_PinMode = GPIO_MODE_OUT;
```

```
    Gpioled.GPIO_PinConfig.GPIO_PinSpeed = GPIO_SPEED_FAST;
```

```
    Gpioled.GPIO_PinConfig.GPIO_PinOPtype = GPIO_OP_TYPE_PP; // push pull config
```

```
    Gpioled.GPIO_PinConfig.GPIO_PinPuPdControl = GPIO_NO_PUPD; // no enabling  
    because already it is push pull
```

```
    GPIO_PerioClockControl(GPIOD, ENABLE); // enabling peripheral clock due to lot of  
    peripherals are required to control
```

```
    GPIO_Init(&Gpioled);
```

```
    /* Toggle led */
```

```
    while(1){
```

```
        GPIO_ToggleOutputPin(GPIOD,GPIO_PIN_NO_12);
```

```
        delay();
```

```
    }
```

```
    //GPIO_Init(&Gpioled);
```

```
    return 0;
```

```
}
```

2) LED TOGGLE USING OPEN DRAIN CONFIGURATION FOR THE OUTPUT PIN

```
Gpioled.GPIO_PinConfig.GPIO_PinOPtype = GPIO_OP_TYPE_OD; // push pull config
```

```
Gpioled.GPIO_PinConfig.GPIO_PinPuPdControl = GPIO_NO_PUPD; // no enabling because  
already it is push pull
```

For the above program the led will not blink.

OR

```
Gpioled.GPIO_PinConfig.GPIO_PinOPtype = GPIO_OP_TYPE_OD; // push pull config
```

```
Gpioled.GPIO_PinConfig.GPIO_PinPuPdControl = GPIO_PIN_PU;
```

For the above program LED will blink with very low intensity

Another way to blink LED is Take jumper wire and make connections from PD12 to VCC 5volts on board connect through a resistor this can be done by GPIO_NO_PUPD

3) LED TOGGLE USING INTERNAL BUTTON

```
#include "stm32f407xx.h"
```

```
#define HIGH 1
```

```
#define BTN_PRESSED HIGH
```

```
void delay(void){
```

```
    for(uint32_t i = 0; i < 500000/2 ; i++);
```

```
}
```

```
int main(void){
```

```
    GPIO_Handle_t Gpioled,GPIObtn;
```

```
    Gpioled.pGPIOx = GPIOD; //pd12 led4 so port d
```

```
    Gpioled.GPIO_PinConfig.GPIO_PinNumber = GPIO_PIN_NO_12;
```

```
Gpioled.GPIO_PinConfig.GPIO_PinMode = GPIO_MODE_OUT;
Gpioled.GPIO_PinConfig.GPIO_PinSpeed = GPIO_SPEED_FAST;
Gpioled.GPIO_PinConfig.GPIO_PinOPtype = GPIO_OP_TYPE_PP; // push pull config
Gpioled.GPIO_PinConfig.GPIO_PinPuPdControl = GPIO_NO_PUPD; // no enabling
because already it is push pull
```

GPIO_PeriClockControl(GPIOD, ENABLE); // enabling peripheral clock due to lot of peripherals are required to control

GPIO_Init(&Gpioled);

```
GPIOBtn.pGPIOx = GPIOA; //PA0 B1 USER so A port 0 pin
GPIOBtn.GPIO_PinConfig.GPIO_PinNumber = GPIO_PIN_NO_0;
GPIOBtn.GPIO_PinConfig.GPIO_PinMode = GPIO_MODE_IN;
GPIOBtn.GPIO_PinConfig.GPIO_PinSpeed = GPIO_SPEED_FAST;
//GPIOBtn.GPIO_PinConfig.GPIO_PinOPtype = GPIO_OP_TYPE_OD; does not apply
because only when mode is output it is applicable
GPIOBtn.GPIO_PinConfig.GPIO_PinPuPdControl = GPIO_NO_PUPD; // no enabling
because already it is push pull
```

GPIO_PeriClockControl(GPIOA, ENABLE); // enabling peripheral clock due to lot of peripherals are required to control

GPIO_Init(&GPIOBtn);

/* Toggle led */

```
while(1){
    if(GPIO_ReadFromInputPin(GPIOA, GPIO_PIN_NO_0) == BTN_PRESSED)
    {

        delay();
    }
}
```

```

        GPIO_ToggleOutputPin(GPIOD,GPIO_PIN_NO_12);

    }

}

//GPIO_Init(&Gpioled);

return 0;
}

```

4) TOGGLE LED USING EXTERNAL BUTTON AND LED

```

#include "stm32f407xx.h"

#define HIGH 1
#define LOW 0
#define BTN_PRESSED HIGH // LOW

void delay(void){
    for(uint32_t i = 0; i < 500000/2 ; i++);
}

int main(void){

    GPIO_Handle_t Gpioled,GPIOBtn;
    Gpioled.pGPIOx = GPIOA;
    Gpioled.GPIO_PinConfig.GPIO_PinNumber = GPIO_PIN_NO_8;
    Gpioled.GPIO_PinConfig.GPIO_PinMode = GPIO_MODE_OUT;
    Gpioled.GPIO_PinConfig.GPIO_PinSpeed = GPIO_SPEED_FAST;
    Gpioled.GPIO_PinConfig.GPIO_PinOType = GPIO_OP_TYPE_PP; // push pull config

```

Gpioled.GPIO_PinConfig.GPIO_PinPuPdControl = GPIO_NO_PUPD; // no enabling because already it is push pull

GPIO_PerioClockControl(GPIOA, ENABLE); // enabling peripheral clock due to lot of peripherals are required to control

GPIO_Init(&Gpioled);

GPIOBtn.pGPIOx = GPIOB;

GPIOBtn.GPIO_PinConfig.GPIO_PinNumber = GPIO_PIN_NO_12;

GPIOBtn.GPIO_PinConfig.GPIO_PinMode = GPIO_MODE_IN;

GPIOBtn.GPIO_PinConfig.GPIO_PinSpeed = GPIO_SPEED_FAST;

//GPIOBtn.GPIO_PinConfig.GPIO_PinOPtype = GPIO_OP_TYPE_OD; does not apply because only when mode is output it is applicable

GPIOBtn.GPIO_PinConfig.GPIO_PinPuPdControl = GPIO_PIN_PU; // Enabling push pull

GPIO_PerioClockControl(GPIOA, ENABLE); // enabling peripheral clock due to lot of peripherals are required to control

GPIO_Init(&GPIOBtn);

/* Toggle led */

```
while(1){  
    if(GPIO_ReadFromInputPin(GPIOB, GPIO_PIN_NO_12) == BTN_PRESSED)  
    {  
  
        delay();  
        GPIO_ToggleOutputPin(GPIOA,GPIO_PIN_NO_8);  
  
    }  
}
```

```
}
```

```
//GPIO_Init(&Gpioled);
```

```
return 0;
```

```
}
```